Web Scraping Exercise: DataFrames to CSV

Explore how to transform web-scraped HTML tables into pandas DataFrames and export these as CSV files. This lesson helps you practice data extraction techniques needed for ETL processes, enabling efficient preparation of data for analysis and further processing.

We've already read and parsed a Wikipedia web page containing tables ranking the world's largest banks by total assets and by country. We'll leverage this parsed data to create pandas DataFrames and export them as CSV files for further analysis.

Let's convert the parsed HTML tables into pandas DataFrames and export them as CSV files.

Create DataFrames using pandas

After collecting the wikitables from Wikipedia, import `pandas` and use the `pd.read_html()` method to convert these HTML tables into a list of DataFrame objects.

main.py Python 3.10.4

```
# Read HTML content into a list of pandas DataFrame
print('\nConverting HTML tables into Pandas DataFrames\n')
tables_list = pd.read_html(str(wiki_tables))
print(tables_list)
```

Ask

Run

Process tables

Print the DataFrames

Next, extract the three DataFrames from the list `tables_list` into their own separate objects. The DataFrames are located at the first, second, and third indexes in the list.

main.py

Python 3.10.4

```
# Extract DataFrames from the list: tables_list
banksByTotalAssets = <<InsertHere>>
banksByCountry = <<InsertHere>>

# Print DataFrames
print(banksByTotalAssets)
print(banksByCountry)
```

Run

Extract DataFrames from a list

Ask

Export DataFrames as CSV

Lastly, export the three DataFrames as CSV files for later analysis using the `pandas` DataFrame method `to_csv()`.

main.py Python 3.10.4

```
# Export DataFrames
print('\nExporting as csv...')
banksByTotalAssets.<<InsertHere>>('Banks_By_Assets.csv', index = False)
banksByCountry.<<InsertHere>>('Banks_By_Country.csv', index = False)

# Verify by reading the first 10 rows of 'Banks_By_Assets.csv'
<<InsertHere>>
```

Run

Export DataFrames as CSV

Complete solution code

Here's the complete solution for the above tasks:

main.py Python 3.10.4

```
# Import
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = 'https://en.wikipedia.org/wiki/List_of_largest_banks'

# Sending a GET request
response = requests.get(url)

# Check status code and print a message accordingly
if response.status_code == 200:
```

Ask

```
print('HTTP request is sucessful \n')
```

Run

Solution

- **Line 4:** We import the `pandas` library as `pd`.
- **Line 31:** We use `pd.read_html()` to create a list of two DataFrames out of the `wiki_tables`.
- **Line 34:** We extract the table "Banks By Total Asset" from the first index in the `tables_list` list.
- **Line 35:** We extract the table "Banks By Country" from the second index in the `tables_list` list.
- **Lines 39–40:** We use `to_csv(FileName, index = False)` to export the three DataFrames as CSV files.
- **Line 43:** We verify that the operation is successful by reading the first ten rows of `Banks_By_Assets.csv`.

Ask